



Lightweight single image super-resolution based on multi-path progressive feature fusion and attention mechanism

Shanshan Li¹ · Dengwen Zhou¹ · Yukai Liu¹ · Dandan Gao¹ · Wanjun Wang¹

Received: 18 August 2022 / Accepted: 16 April 2023 / Published online: 17 May 2023
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2023

Abstract

Single image super-resolution (SISR) is a fundamental image processing task, which aims to generate a high-resolution (HR) image from its low-resolution (LR) counterpart. Deep convolution neural networks (CNNs) have significantly improved the performance of SISR, and dominated the current research on SISR techniques. However, the performance improvement depends heavily on the size of the networks. In general, the deeper the networks, the better the performance, which limits their use to devices with limited computing and memory resources. The challenge is to find the optimal balance between network model complexity and SISR performance. In this paper, we propose a new lightweight SISR algorithm based on CNNs, which uses multi-path progressive feature fusion and attention mechanism. Our main contributions are as follows: (1) We propose a multi-path progressive feature fusion block (PFF), which can use the feature from the previous path to gradually guide the feature learning of the next path step by step in multiple paths. (2) We propose a multi-path feature attention mechanism (FAM), which can adaptively weigh the multi-path feature channels to be concatenated, improve the utilization of feature information and feature representation capability. The experimental results show that whether it is an objective measurement or a subjective measurement, our method is better than other similar state of the art methods, and has a better model complexity and performance balance.

Keywords Image super-resolution · Convolutional neural network · Feature fusion · Attention mechanism

1 Introduction

Single image super-resolution (SISR) [8] is a fundamental image processing task, which aims to generate a high-resolution (HR) image from its low-resolution (LR) counterpart, and has a wide range of applications in remote sensing [41], video surveillance [50], medical imaging [37] and so on.

However, SISR is difficult, because it is an inherently ill-posed inverse problem. One LR image can correspond to multiple HR images [36] and the prior knowledge is needed to regularize the solution space.

SISR has a long history of research [34]. In recent years, deep convolutional neural networks (CNNs) have dominated the research of SISR methods [6, 7, 20–22, 39, 56], which learn mapping functions between LR and HR images directly end-to-end, and significantly improve the performance of SISR. However, CNNs-based SISR methods are heavily dependent on the size of the networks, i.e., the number of parameters, depth (number of layers) and width (number of channels) in the networks. To improve the performance of SISR, larger networks are often required. For example, EDSR [27] has 65 convolutional layers and about 43 M parameters; RCAN [55] has over 800 convolutional layers and about 16 M parameters. Although methods such as EDSR and RCAN have good performance, they require high computing and storage cost and are difficult to apply to resource-constrained devices (e.g., mobile phones). Designing lightweight networks (with low computing and storage

✉ Dengwen Zhou
zdw@ncepu.edu.cn
Shanshan Li
lss@ncepu.edu.cn
Yukai Liu
liuyk@ncepu.edu.cn
Dandan Gao
gdd@ncepu.edu.cn
Wanjun Wang
1206371055@qq.com

¹ School of Control and Computer Engineering, North China Electric Power University, Beijing 102206, People's Republic of China

requirements) is a hot topic of current SISR research. The challenge is to establish a better balance between the model complexity and the performance of the SISR methods.

Multi-path SISR network architectures are very popular. Zhao et al. [57] propose a very lightweight SISR network, called the pixel attention network (PAN). The basic building block of PAN (i.e., SC-PA block) contains two branches: one branch contains pixel attention (PA), which is similar to channel attention [14] and spatial attention [47] for higher-level feature manipulation. The other branch uses a standard convolutional layer to maintain the original information. Then, the output feature channels of the two branches are concatenated along the channel dimension. The research of Wang et al. [44] shows that the features of the previous layers (i.e., auxiliary features) can help the learning of the features of the current layer. The basic building block of their proposed A²F model is the A²F block, which also consists of two main branches: one branch uses the features of all previous blocks to compute the channel attention, the other branch computes the feature using the WDSR A block [49], and then the features of both branches and the original feature are weighted and concatenated together. The research of Zhao et al. [58] shows that the weighted channel concatenation is more in line with the behavior of human brain neurons and can improve the expressive ability of the features. Inspired mainly by the works of Wang et al. [44], Zhao et al. [57, 58], we propose a lightweight SISR network based on multi-path progressive feature fusion and attention mechanism (MPFFA). The core building block of MPFFA is the multi-path feature extraction block (MPFE), as shown in Fig. 2d. MPFE includes two sub-blocks: multi-path progressive feature fusion (PFF) and multi-path feature attention mechanism (FAM). Our MPFE is somewhat more similar to A²F, but, unlike A²F, the PFF uses the feature from the previous branch in the current block to gradually guide the feature learning of the next branch to predict the importance of the features pixel by pixel, without using the features from the previous blocks, as in A²F. In addition, when FAM concatenates the features of each branch along the channel dimension, the weights of the feature channels in each branch are computed by the channel attention mechanism, instead of using a scalar weight, which can more accurately weight the feature information. Experimental results show that our MPFFA is very effective and outperforms the other state-of-the-art lightweight SISR methods, taking the model complexity and performance balance to a new level (please refer to Fig. 1).

In summary, our contributions include: (1) We propose a novel lightweight SISR network (MPFFA) based on multi-path progressive feature fusion and attention mechanism, which outperforms the current state of the art lightweight SISR methods with better model complexity and performance balance; (2) We propose a multi-path progressive

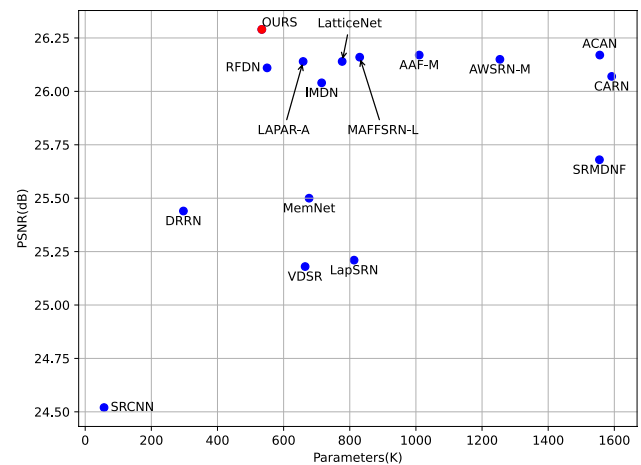


Fig. 1 Average PSNRs and numbers of parameters for 4× SR models for each state of the art SISR method on the Urban100 dataset. The horizontal and vertical axes represent the number of parameters (K) and PSNR (dB), respectively. The red dot represents our MPFFA, and the blue dots are the other methods

feature fusion block (PFF), which can use the previous features to guide the learning of subsequent multi-path features gradually; (3) We propose a multi-path feature attention mechanism (FAM), which can better weight and concatenate the multi-path features, and thus enhance the expression ability of features.

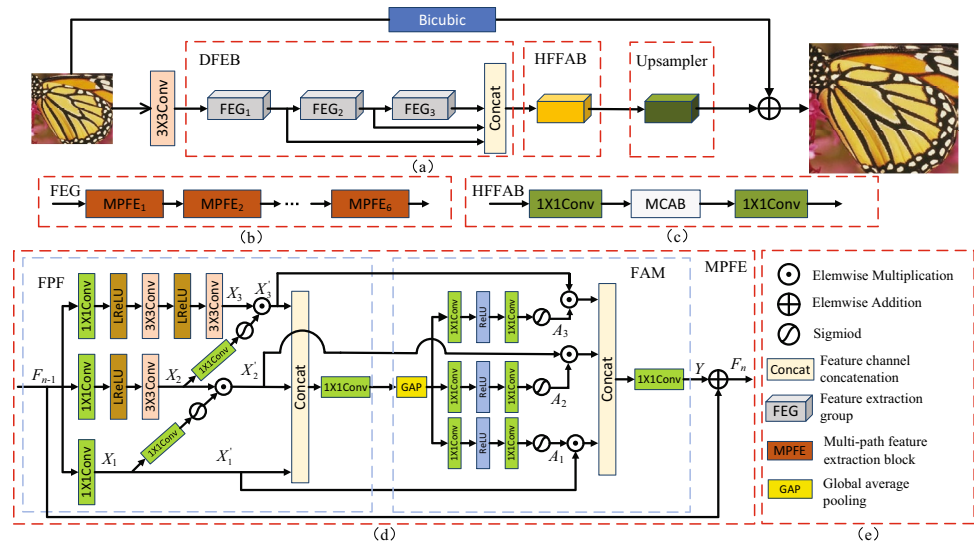
The remainder of this paper is organized as follows. In Sect. 2 the related works are presented. Section 3 describes the proposed methods. Section 4 reports experimental results and Sect. 5 concludes the paper.

2 Related works

2.1 Lightweight SISR networks

Dong et al. [6] propose the seminal CNNs-based SISR method, called SRCNN. Recent years have witnessed a surge of CNNs-based SISR methods [6, 7, 20–22, 39, 56]. Currently, CNNs have dominated the research of SISR methods. The CNNs-based SISR methods usually use L1 or L2 loss, and one of its problems is that the generated SR image is too smooth and lacks high-frequency details [25]. Generative Adversarial Networks (GANs) [10] are also introduced into SISR [25], which use perceptual and adversarial losses. GANs-based SISR methods [25, 59] can recover clearer high-frequency details, however, the generated SR images have low fidelity, severe distortion, and are also difficult to train. One of the other major problems of CNNs-based SISR methods is the heavy dependence on the size of the networks. The larger the network size, the stronger the expressive power and the

Fig. 2 **a** The network architecture of our proposed MPFFA. **b** Feature extraction group (FEG). **c** Hierarchical feature fusion attention block (HFFAB). **d** Multi-path feature extraction block (MPFE). **e** Sign description



better the performance, which also limits their practicality, especially for devices with limited computational and memory resources. Lightweight SISR methods based on CNNs, have attracted widespread attention. The challenge is to find the best balance between model complexity and performance. In lightweight SISR network structure design, the techniques of recursion are very popular. The DRCN proposed by Kim et al. [21] introduces recursion into the CNNs-based SISR methods for the first time. The parameters are shared among the convolutional layers, which can significantly reduce the model parameters. Tai et al. [39] propose a very deep persistent memory network (MemNet), which uses residual blocks as recursive units. The ACAN proposed by Zhou et al. [60] also uses residual blocks as recursive units. Although the recursive structure can reduce parameters, it cannot reduce the computational cost. 1×1 convolution, group convolution [3], depthwise separable convolution [13], and self-calibrating convolution [29], etc. are also widely used in lightweight SISR network architectures. Zhao et al. [57] propose a very lightweight SISR model, called PAN, which uses self-calibration convolutions as the basic building blocks, and also uses 1×1 convolution to design a lightweight pixel attention block. The basic building block of the IDN proposed by Hui et al. [18] is the information distillation block, which uses 1×1 convolution to compress and distill feature information. Hui et al. [19] also propose the information multi-distillation network (IMDN), which uses the information multi-distillation block (IMDB) as the basic building blocks. IMDB improves DBlock, using multi-level channel splitting, and processive refinement. The RFDN proposed by Liu et al. [28] further improves the channel splitting and feature distillation of IMDN to make it more lightweight and more effective. In this paper, our model adopts a novel multi-path progressive feature

fusion attention mechanism, which significantly improves the complexity and performance balance of current SISR models.

2.2 Multi-pathing learning

Multi-path learning is an important paradigm of the CNNs-based SISR network architectures. Residual connections [20] and dense connections [15] are probably the most common multi-path learning architectures. They can enhance the propagation of feature information, but also suffer from serious feature information redundancy. How to improve multi-path learning and increase the utilization and expressive ability of feature information is an important concern for the SISR research community [11, 17, 19, 28, 44, 57]. Han et al. [11] propose a dual-state recurrent network (DSRN), which extracts features simultaneously in LR and HR spaces through two branches and continuously interacts with each other to enhance the expression ability of features. Hui et al. [19] propose an information multi-distillation block (IMDB) which progressively extracts and refines the hierarchical features by channel splitting, with one branch maintaining the original feature information and the other branch performing convolutional operations. Liu et al. [28] propose a RFDB block, using the feature distillation connection (FDC), instead of the channel splitting in the IMDB block, which is more lightweight and more flexible. Huang et al. [17] propose a dual-path attention network (DPAN), the basic building block of which is the dual-path block (DPB), in which one branch performs residual learning and the other branch performs channel concatenation to better fuse the complementary feature information. The attentive auxiliary feature block (A²F) proposed by Wang et al. [44] contains two main branches: one branch projects the features of the preceding layers; the other branch uses the channel attention

mechanism, which learns the importance of each channel. Zhao et al. [57] propose a pixel attention network (PAN), the basic building block of which is the self-calibrated block with pixel attention (SC-PA). The SC-PA block consists of two branches: one branch uses pixel-attention (PA) to perform high-level feature manipulation, and the other branch uses standard convolutional layers to maintain the original information. Our model also uses a multi-path (branch) structure, with the main innovation being that the feature from one branch is used to progressively guide and calibrate the feature learning in the next branch.

2.3 Attention mechanism

Attention mechanism [4] usually refers to the human visual system, which can automatically focus on the salient areas. The attention mechanism in CNN network architectures is similar, which means that it can focus on specific parts of the input, which can improve the utilization and expressive ability of features. Hu et al. [14] propose a squeeze-and-excitation network (SENet), where the basic building block is squeeze-and-excitation (SE) blocks. The SE block uses global average pooling (GAP) to squeeze each input channel into channel descriptors (constants), which are then fed into two fully-connected layers to generate the weights for each input channel to explicitly model the interdependencies between channels. The residual channel attention network (RCAN) proposed by Zhang et al. [55] introduces the SENet mechanism [14] of Hu et al. into SISR. Dai et al. [5] further propose a second-order channel attention block, which adaptively rescales each feature channel by using second-order feature statistics, instead of GAP, and can better learn the correlations between feature channels. Woo et al. [47] introduce both channel attention and spatial attention mechanisms. It is possible to focus not only on the feature channels, but also on the spatial positions in the channels. Zhang et al. [52] propose a multi-context attentive block (MCAB), which can also focus on the dependencies between channel and spatial dimensions at the same time. In this paper, we propose a multi-path attention mechanism that can adaptively weight the multi-path features. We also use MCAB [52] to fuse the hierarchical features from the model.

3 Proposed method

The network architecture of our MPFFA is shown in Fig. 2a. It mainly includes four parts: Shallow feature extraction block (SFEB), Deep feature extraction block (DFEB), Hierarchical feature fusion attention block (HFFAB) and Upsampler. SFEB only has a 3×3 convolutional layer to extract shallow features. Upsampler

uses the upsampling block of PAN [57]. HFFAB mainly uses the multi-context attentive block (MCAB) in TSAN [52], as shown in Fig. 2c. To reduce the computational effort, the number of feature channels input to MCAB is compressed four times by 1×1 convolution, while the number of feature channels output from MCAB is expanded four times by 1×1 convolution. Our work is primarily focused on DFEB, which includes three feature extraction groups (FEG). Each FEG contains six multi-path feature extraction blocks (MPFE), as shown in Fig. 2b. Each MPFE consists of a multi-path progressive feature fusion block (PFF) and a multi-path feature attention mechanism (FAM), as shown in Fig. 2d. MPFFA also contains a global residual connection, where the LR image is enlarged to the size of the target SR image by bicubic interpolation.

Assuming that the input LR image is I_{LR} , it is first fed into the SFEB block to extract the shallow feature:

$$F_0 = H_{SFEB}(I_{LR}) \quad (1)$$

where $H_{SFEB}(\cdot)$ is the SFEB function, and F_0 is its output feature. F_0 is then fed into the DFEB to further extract the deep features. DFEB contains three FEGs, and each FEG contains six MPFEs. Therefore, F_0 passes through eighteen MPFEs, and the output of each MPFE can be expressed as:

$$F_n = H_n(F_{n-1}), (n = 1, 2, \dots, 18) \quad (2)$$

where $H_n(\cdot)$ is the n -th MPFE function, and F_n is its output feature. Considering that hierarchical features can provide complementary feature information [60], the hierarchical features F_6 , F_{12} and F_{18} output by the three FEGs are concatenated to improve feature utilization and expressiveness:

$$F_{cat} = [F_6, F_{12}, F_{18}] \quad (3)$$

where $[\cdot]$ is the feature channel concatenation operation, and F_{cat} is the concatenated feature, which is also the output feature of the DFEB block, and then is fed into HFFAB:

$$F_{hff} = H_{HFFAB}(F_{cat}) \quad (4)$$

where $H_{HFFAB}(\cdot)$ is the HFFAB function, and F_{hff} is its output feature, which is then fed into Upsampler block:

$$I_{SR} = H_{Up}(F_{hff}) + H_{Bic}(I_{LR}) \quad (5)$$

where $H_{Up}(\cdot)$ is the Upsampler function, and $H_{Bic}(\cdot)$ is the bicubic interpolation function, which is added to the output of the residual $H_{Up}(\cdot)$ to obtain the final output of the target SR image I_{SR} . In the current SISR methods, most upsampling blocks use sub-pixel convolution [27] or transposed convolution [9], here we use the upsampling block of PAN [57].

3.1 Multi-path progressive feature fusion block (PFF)

The multi-path progressive feature fusion block (PFF), mainly inspired by the works of Wang et al. [44] and Zhao et al. [57], performs multi-path (we use three paths) progressive feature fusion, which uses the feature of the previous branch to guide and calibrate the feature learning of the next branch. As shown on the left side of Fig. 2d, the bottom branch contains only a 1×1 convolutional layer. The middle branch contains a 1×1 convolutional layer and a 3×3 convolutional layer, with an LReLU [12] nonlinear activation function between the two convolutional layers. The feature of the bottom branch guides and calibrates the feature of the middle branch by the 1×1 convolution and Sigmoid function. The top branch contains a 1×1 convolutional layer and two 3×3 convolutional layers, and there is an LReLU nonlinear activation function between every two convolutional layers. The feature of the middle branch guides and calibrates the feature of the top branch by the 1×1 convolution and the Sigmoid function again. Considering that F_{n-1} and F_n are the input and output of the n -th MPFE, respectively, as shown in Eq. (2). The output features of the three branches before calibration are as follows (ignoring the LReLU nonlinear activation functions):

$$X_1 = \text{Conv}_{1 \times 1}(F_{n-1}) \quad (6)$$

$$X_2 = \text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F_{n-1})) \quad (7)$$

$$X_3 = \text{Conv}_{3 \times 3}(\text{Conv}_{3 \times 3}(\text{Conv}_{1 \times 1}(F_{n-1}))) \quad (8)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, and $\text{Conv}_{3 \times 3}(\cdot)$ is the 3×3 convolution function. The output features of the three branches after calibration can be described as:

$$X'_1 = X_1 \quad (9)$$

$$X'_2 = X_2 \odot (\sigma(\text{Conv}_{1 \times 1}(X_1))) \quad (10)$$

$$X'_3 = X_3 \odot (\sigma(\text{Conv}_{1 \times 1}(X_2))) \quad (11)$$

where $\sigma(\cdot)$ is the Sigmoid function, and the $\odot$ is the element-wise multiplication. Then, the calibrated output feature channels of the three branches are concatenated, and the channels are compressed and fused with a 1×1 convolutional layer:

$$X' = \text{Conv}_{1 \times 1}([X'_1, X'_2, X'_3]) \quad (12)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, and $[\cdot]$ is the feature channel concatenation operation.

3.2 Multi-path feature attention mechanism (FAM)

The multi-path feature attention mechanism (FAM) is mainly inspired by the works of Wang et al. [44] and Zhao et al. [57], which indicate that the weighted concatenation of feature channels can improve the expressive ability of features. Unlike their works, instead of using scalar weights, the FAM uses the channel attention mechanism to optimally weight the channels of the features from each branch of the PFF and then concatenate them. The output feature X' of the n -th PFF block, through global average pooling (GAP), generates an initial weight vector. Then, the weight vector is processed by three branches. The feature channels of each branch are compressed and expanded by two 1×1 convolutions (with a ReLU [1] nonlinear activation function between the two 1×1 convolutions). Next, using the Sigmoid function generates the weight vector of the feature channels of each branch.

Assuming the concatenated feature output by the n -th PFF block $X' = [x_1, \dots, x_c, \dots, x_C] \in \mathbb{R}^{H \times W \times C}$ (H , W and C are the height, width and number of channels of the feature, respectively), the GAP can be expressed as:

$$z_c = H_{GAP}(x_c) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W x_c(i, j) \quad (13)$$

where $H_{GAP}(\cdot)$ is the GAP function, and $Z = [z_1, \dots, z_c, \dots, z_C] \in \mathbb{R}^C$. The weight of each channel of the three branches can be expressed as (ignoring the ReLU nonlinear activation function):

$$A_k = \sigma(\text{Conv}_{1 \times 1}(\text{Conv}_{1 \times 1}(Z))), (k = 1, 2, 3) \quad (14)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, and $\sigma(\cdot)$ is the Sigmoid function. The residual output Y of the n -th FAM can be expressed as:

$$Y = \text{Conv}_{1 \times 1}([A_1 \odot X'_1, A_2 \odot X'_2, A_3 \odot X'_3]) \quad (15)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, $X'_k (k = 1, 2, 3)$ is the output feature of the k -th branch of the n -th PFF block, $[\cdot]$ is the feature channel concatenation operation, and $\odot$ is the element-wise multiplication. Since the residual connection is used, the output feature F_n of the n -th FAM needs to be added to the input F_{n-1} of the n -th PFF block:

$$F_n = Y + F_{n-1} \quad (16)$$

3.3 Hierarchical feature fusion attention block (HFFAB)

The multi-context attentive block (MCAB) [52] can focus on information-rich contextual features. The hierarchical

feature fusion attention block (HFFAB) only adds the 1×1 convolutional layers before and after the MCAB, as shown in Fig. 2(c). The preceding 1×1 convolutional layer compresses the feature channels to reduce the computational effort; the following 1×1 convolutional layer then expands the feature channels. The operation of HFFAB block, see Eq. (4), can be further expressed as:

$$F_{hff} = \text{Conv}_{1 \times 1}(H_{MCAB}(\text{Conv}_{1 \times 1}(F_{cat}))) \quad (17)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is the 1×1 convolution function, $H_{MCAB}(\cdot)$ is the MCAB block function, and F_{cat} is the output feature of the DFEB block.

4 Experiments

4.1 Settings

We used the popular DIV2K [42] dataset as the training and validation data. The first 800 images (001–800) were used for training, and 10 images (821–830) were used for validation, labeled as DIV2K_val10. The original HR training images were bicubically downsampled to obtain the paired LR training images. Similar to other methods, the training images were augmented by randomly rotating 90° , 180° , 270° and flipping horizontally. Our test datasets were five standard benchmark ones, Set5 [3], Set14 [51], B100 [31], Urban100 [16], and Manga109 [32]. As with the previous methods, the SR results were evaluated by PSNR and SSIM [46] metrics on the luminance (Y) channel of the transformed YCbCr space [56]. We first trained the $2\times$ model and then train the $3\times$ and $4\times$ models based on the pre-trained $2\times$ model.

In the model training, 32 patches with the size of 48×48 were randomly selected for each batch. The Adam [23] optimizer was used, and set $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. The model was trained for 1200 epochs with an initial learning rate set to 2×10^{-4} and halved for every 200 epochs. The L1 loss function was used. The model was implemented by using PyTorch [35] with a NVIDIA RTX 2080Ti GPU.

4.2 Ablation study

In order to evaluate the effects of the multi-path progressive feature fusion block (PFF), the multi-path feature attention mechanism (FAM) and the hierarchical feature fusion attention block (HFFAB), we conducted four comparative experiments on DIV2K_val10 validation dataset. The first experiment did not include FAM and HFFAB blocks. The PFF block also contained only three branches, and did not include the calibration of the previous features to the subsequent features, that is, it did not include the 1×1 convolution and Sigmoid function between the two branches. This experiment was called MPFFA_0. In the second experiment, the calibration of the previous features to the subsequent features was added to MPFFA_0. That is, the PFF block was completely included. However, the FAM and HFFAB blocks were still not included. This experiment was called MPFFA_1. The third experiment was to add the HFFAB block to MPFFA_1, which was called MPFFA_2. The fourth experiment was to add the FAM block to MPFFA_2, which was our complete MPFFA network model. The PSNRs of the four models on the DIV2K_val10 validation dataset with $3\times$ SR were shown in Table 1. It could be seen that the PSNR of MPFFA_0 was only 29.49 dB and the PSNR of MPFFA_1 was 0.11 dB higher than MPFFA_0. It seemed to indicate that the multi-path progressive feature fusion, which used the previous features assisting the learning of the subsequent features, was effective. Compared with MPFFA_1, the PSNR of MPFFA_2 also increased by 0.03 dB. Compared with MPFFA_2, the PSNR of the final model MPFFA increased by 0.06 dB, which indicated that the FAM block using the weighted concatenation of multi-path feature channels was also effective.

4.3 Experiments on the number of MPFEs

To ensure that our model was effective enough and light-weight enough, three feature extraction groups (FEGs) were used, and each FEG contained six multi-path feature extraction blocks (MPFE). The selection of the number of MPFEs needed to consider the impacts on the number of parameters, the computational effort, and the performance. We conducted three experiments on the DIV2K_val10 validation

Table 1 Average PSNRs, number of parameters and computational cost of $3\times$ SR on validation dataset DIV2K_val10 for different variants of network architecture

Model	PFF's calibration	PFF's three branches	HFFAB	FAM	PSNR (dB)	Parameters (K)	FLOPs (G)
MPFFA_0	✓	×	×	×	29.49	443	59.4
MPFFA_1	✓	✓	×	×	29.60	465	61.6
MPFFA_2	✓	✓	✓	×	29.63	480	63.0
MPFFA	✓	✓	✓	✓	29.69	523	66.2

dataset. In each experiment, the number of MPFEs in each FEG was set to 5, 6 and 7, respectively. The PSNR results for 3× SR were shown in Table 2. It could be seen that increasing the number of MPFEs could improve the performance of the model, but the number of parameters also increased. It was assumed that the number of MPFEs was represented by n . Compared to $n = 5$, when $n = 6$, the number of parameters increased by 78 K and the PSNR increased by 0.14 dB. Compared to $n = 7$, when $n = 6$, the number of parameters was 78 K less and the PSNR reduced by only 0.05 dB. $n = 6$ was probably a reasonable compromise.

4.4 Experiments on FAM

4.4.1 Experiments on the number of branches

To gain further insight into the effect of multi-paths in the FAM, we experimented with the case of the FAM with only one branch on the DIV2K_val10 validation dataset. The PSNR of the 3× SR was 29.56 dB, which was 0.13 dB lower than the PSNR when the FAM contained three branches. This result was interesting because when the FAM contained only one branch, the PSNR was 0.07 dB lower than the that of MPFFA_2 (without FAM block).

4.4.2 Experiments with shared parameters for each branch

Parameter sharing can reduce the number of parameters. If the parameters of each branch were shared in the FAM, the PSNR of 3× SR was 29.65 dB on the validation dataset DIV2K_val10, which was 0.04 dB lower than that of the case where the parameters of each branch were not shared.

4.5 Comparisons with state-of-the-arts

In order to verify the effectiveness of our MPFFA method, we conducted objective quantitative comparisons and subjective visual effect comparisons with the other state-of-the-art methods, which included SRCNN[6], FSRCNN [7], VDSR [20], DRCN [21], LapSRN [24], DRRN [40], MemNet [39], SRM-DNF [53], CARN [2], IMDN [19], LAPAR-A [26], RFDN [28], A²F-M [44], ACAN [60], LatticeNet [30], AWSRN-M [43], PAN [57], MAFFSRN-L [33], EMASRN [61] and ShuffleMixer [38]. On five standard test datasets, the average

PSNR/SSIM results for 2×, 3× and 4× SR were shown in Table 3, the visual comparisons were shown in Fig. 3. Except for LatticeNet and PAN, the PSNR/SSIM results of the other methods were obtained from the the original papers. The results of LatticeNet were derived from running the trained model provided by the author [48]. Since the original PAN uses both DIV2K and Flickr2K [45] training datasets, for the fair comparisons, we used the source code provided by the authors and re-trained it on the DIV2K training dataset. Therefore, the test results were slightly different from those in the original paper. Table 3 also included the number of parameters and GFLOPs [54] for each state of the art method. GFLOPs refers to the number of multiplication-addition operations.

Quantitative comparisons: As seen in Table 3, our PSNR/SSIM results were significantly better than the other methods. For example, the cutting-edge LAPAR-A and RFDN were slightly larger than our MPFFA model, but our results were almost always better than theirs. For 2×, 3× and 4× SR on each dataset, our maximum increases in PSNR were 0.31 dB, 0.34 dB, and 0.29 dB for LAPAR-A, and 0.22 dB, 0.18 dB, and 0.18 dB for RFDN, respectively. The number of parameters of A²F-M was about twice ours, but the PSNR results of 2×, 3× and 4× SR on the Manga109 dataset were 0.11 dB, 0.19 dB and 0.14 dB lower than ours, respectively. For 4× SR, the number of parameters of ACAN was about three times higher than ours, but the PSNR on the Manga109 dataset was 0.18 dB lower than ours, while the computational effort of ACAN was about 26 times higher than ours. We also gave the 4× SR inference times of several representative models in Table 4. It was interesting to note that our model had the advantages of PSNR/SSIM performance and computational cost, however, there seemed to be no significant advantage in the inference time, which would be an issue for our further study.

Visual comparisons: Figure 3 showed the 4× SR images of each method. Our results were better than those of other methods, and were significantly visible. Taking the image img067 in Urban100 dataset as an example, the stripes recovered by Bicubic, EMASRN, ShuffleMixer, LatticeNet and A²F-M etc. had severe blurring or distortion. Our results were very close to the original HR image. For the image img089 in the Urban100 dataset, the stripes recovered by other methods were either excessively blurred, or the direction of the stripes was wrong, while we recovered the stripes in the right direction, and also much clearer. The results for the image barbara in the Set14 dataset were also similar.

Table 2 PSNRs and numbers of parameters for 3× SR experiments on DIV2K_val10 for different number of the MPFEs in each FEG

Number of MPFE	PSNR(dB)	Parameters (K)
5	29.55	445
6	29.69	523
7	29.74	601

5 Conclusions

In this paper, we propose a novel lightweight SISR algorithm based on multi-path progressive feature fusion and attention mechanism. Our main work is to propose a multi-path

Table 3 Average PSNR/SSIM results of 2×, 3× and 4× SR for various state-of-the-art SISR methods, with the best and second-best results marked in bold and italics, respectively

Scale	Model	PRMs	FLOPs	Set5	Set14	B100	Urban100	Manga109
×2	SRCNN	57	52.7	36.66/0.9524	32.42/0.9063	31.36/0.8879	29.50/0.8946	35.74/0.9661
	FSRCNN	12	6.0	37.00/0.9558	32.63/0.9088	31.53/0.8920	29.88/0.9020	36.67/0.9694
	VDSR	665	612.6	37.53/0.9587	33.03/0.9124	31.90/0.8960	30.76/0.9140	37.22/0.9729
	DRCN	1774	17974.3	37.63/0.9588	33.04/0.9118	31.85/0.8942	30.75/0.9133	37.63/0.9723
	LapSRN	813	29.9	37.52/0.9590	33.08/0.9130	31.80/0.8950	30.41/0.9100	37.27/0.9740
	DRRN	297	6796.9	37.74/0.9591	33.23/0.9136	32.05/0.8973	31.23/0.9188	37.92/0.9760
	MemNet	677	2662.4	37.78/0.9597	33.28/0.9142	32.08/0.8978	31.31/0.9195	37.72/0.9740
	SRMDNF	1513	347.7	37.79/0.9600	33.32/0.9150	32.05/0.8980	31.33/0.9200	38.07/0.9761
	CARN	1592	222.8	37.76/0.9590	33.52/0.9166	32.09/0.8978	31.92/0.9256	38.36/0.9765
	IMDN	694	158.8	38.00/0.9605	33.63/0.9177	32.19/0.8996	32.17/0.9283	38.88/ 0.9774
	LAPAR-A	548	171.0	38.01/0.9605	33.62/0.9183	32.19/0.8999	32.10/0.9283	38.67/0.9772
	RFDN	534	123.0	38.05/0.9606	33.68/0.9184	32.16/0.8994	32.12/0.9278	38.88/0.9773
	A ² F-M	999	224.2	38.04/0.9607	33.67/0.9184	32.18/0.8996	32.27/0.9294	38.87/ 0.9774
	ACAN	800	2108.0	38.10/0.9608	33.60/0.9177	32.21/0.9001	32.29/0.9297	38.81/0.9773
	LatticeNet	756	169.5	38.06/0.9607	33.70/0.9187	32.20/0.8999	32.25/0.9288	38.94/ 0.9774
	AWSRN-M	1063	244.1	38.04/0.9605	33.66/0.9181	32.21/0.9000	32.23/0.9294	38.66/0.9772
	PAN	261	70.5	37.99/0.9603	33.53/0.9174	32.14/0.8992	31.93/0.9263	38.37/0.9766
	MAFFSRN-L	790	154.4	38.07/0.9607	33.59/0.9177	32.23/0.9005	32.38/0.9308	—/—
	ShuffleMixer	394	91	38.01/0.9606	33.63/0.9180	32.17/0.8995	31.89/0.9257	38.83/ 0.9774
	MPFFA(ours)	523	129.4	38.09/0.9607	33.70/0.9190	32.21/0.9000	32.34/0.9296	38.98/0.9767
×3	SRCNN	57	52.7	32.75/0.9090	29.28/0.8209	28.41/0.7863	26.24/0.7989	30.59/0.9107
	FSRCNN	12	5.0	33.16/0.9140	29.43/0.8242	28.53/0.7910	26.43/0.8080	30.98/0.9212
	VDSR	665	612.6	33.66/0.9213	29.77/0.8314	28.82/0.7976	27.14/0.8279	32.01/0.9310
	DRCN	1774	17974.3	33.82/0.9226	29.76/0.8311	28.80/0.7963	27.15/0.8276	32.31/0.9328
	DRRN	297	67969.0	34.03/0.9244	29.96/0.8349	28.95/0.8004	27.53/0.8378	32.74/0.9390
	MemNet	677	2662.4	34.09/0.9248	30.00/0.8350	28.96/0.8001	27.56/0.8376	32.51/0.9369
	SRMDNF	1530	156.3	34.12/0.9250	30.04/0.8370	28.97/0.8030	27.57/0.8400	33.00/0.9403
	CARN	1592	118.8	34.29/0.9255	30.29/0.8407	29.06/0.8034	27.38/0.8404	33.50/0.9440
	IMDN	703	71.5	34.36/0.9270	30.32/0.8417	29.09/0.8046	28.17/0.8519	33.61/0.9445
	LAPAR-A	594	114.0	34.36/0.9267	30.34/0.8421	29.11/0.8054	28.15/0.8523	33.51/0.9441
	RFDN	541	55.4	34.41/0.9273	30.34/0.8420	29.09/0.8050	28.21/0.8525	33.67/0.9449
	A ² F-M	1003	100.0	34.50/0.9278	30.39/0.8427	29.11/0.8054	28.28/0.8546	33.66/0.9453
	ACAN	1115	1051.7	34.46/0.9277	30.39/ 0.8435	29.11/0.8055	28.28/0.8550	33.61/0.9447
	LatticeNet	765	76.3	34.40/0.9272	30.32/0.8416	29.10/0.8049	28.19/0.8513	33.63/0.9442
	AWSRN-M	1143	116.6	32.42/0.9275	30.32/0.8419	29.13/0.8059	28.26/0.8545	33.64/0.9450
	PAN	261	39.0	34.30/0.9266	30.30/0.8416	29.06/0.8042	28.02/0.8493	33.34/0.9435
	MAFFSRN-L	807	68.5	34.45/0.9277	30.40/0.8432	29.13/0.8061	28.26/ 0.8552	—/—
	EMASRN	427	854.5	34.36/0.9264	30.30/0.8411	29.05/0.8035	28.04/0.8493	33.43/0.9433
	ShuffleMixer	415	43	34.40/0.9272	30.37/0.8423	29.12/0.8051	28.08/0.8498	33.69/0.9448
	MPFFA(ours)	523	66.2	34.51/0.9283	30.37/0.8424	29.13/0.8058	28.35/0.8550	33.85/0.9461

Table 3 (continued)

Scale	Model	PRMs	FLOPs	Set5	Set14	B100	Urban100	Manga109
×4	SRCNN	57	52.7	30.48/0.8628	27.49/0.7503	26.90/0.7101	24.52/0.7221	27.66/0.8505
	FSRCNN	12	4.6	30.71/0.8657	27.59/0.7535	26.98/0.7150	24.62/0.7280	27.90/0.8517
	VDSR	665	612.6	31.35/0.8838	28.01/0.7674	27.29/0.7251	25.18/0.7524	28.83/0.8809
	DRCN	1774	17974.3	31.53/0.8854	28.02/0.7670	27.23/0.7233	25.14/0.7510	28.98/0.8816
	LapSRN	813	149.4	31.54/0.8850	28.19/0.7720	27.32/0.7280	25.21/0.7560	29.09/0.8845
	DRRN	297	6796.9	31.68/0.8888	28.21/0.7720	27.38/0.7284	25.44/0.7638	29.46/0.8960
	MemNet	677	2662.4	31.74/0.8893	28.26/0.7723	27.40/0.7281	25.50/0.7630	29.42/0.8942
	SRMDNF	1555	89.3	31.96/0.8930	28.35/0.7770	27.49/0.7340	25.68/0.7730	30.09/0.9024
	CARN	1592	90.9	32.13/0.8937	28.60/0.7806	27.58/0.7349	26.07/0.7837	30.47/0.9084
	IMDN	715	40.9	32.21/0.8948	28.58/0.7811	27.56/0.7353	26.04/0.7838	30.45/0.9075
	LAPAR-A	659	94.0	32.15/0.8944	28.61/0.7818	27.61/0.7366	26.14/0.7871	30.42/0.9074
	RFDN	550	31.6	32.24/0.8952	28.61/0.7819	27.57/0.7360	26.11/0.7858	30.58/0.9089
	A ² F-M	1010	56.7	32.28/0.8955	28.62/0.7828	27.58/0.7364	26.17/0.7892	30.57/0.9100
	ACAN	1556	616.5	32.24/0.8955	28.62/0.7824	27.59/0.7366	26.17/0.7891	30.53/0.9086
	LatticeNet	777	43.6	32.18/0.8943	28.61/0.7812	27.57/0.7355	26.14/0.7844	30.54/0.9075
	AWSRN-M	1254	72.0	32.21/0.8954	28.65/0.7832	27.60/0.7368	26.15/0.7884	30.56/0.9093
	PAN	272	28.2	32.06/0.8939	28.56/0.7813	27.55/0.7352	26.01/0.7830	30.33/0.9069
	MAFFSRN-L	830	38.6	32.20/0.8953	28.62/0.7822	27.59/0.7370	26.16/0.7887	-/-
	EMASRN	546	1054.4	32.17/0.8948	28.57/0.7809	27.55/0.7351	26.01/0.7838	30.41/0.9076
	ShuffleMixer	411	28	32.21/0.8953	28.66/0.7827	27.61/0.7366	26.08/0.7835	30.65/0.9093
	MPFFA(ours)	534	43.0	32.36/0.8968	28.69/0.7835	27.61/0.7372	26.29/0.7902	30.71/0.9115

The number of parameters(K) and FLOPs(G) for each method were also included

Table 4 Inference times of 4× SR for different models (Low-resolution images were all super-resolved to 1280×720 size)

Model	Time (ms)
VDSR	131.78
DRRN	363.29
MemNet	45.70
CARN	26.44
LAPAR-A	33.34
RFDN	10.29
A ² F-M	17.86
ACAN	117.51
LatticeNet	23.17
AWSRN-M	17.78
EMASRN	525.57
ShuffleMixer	28.56
MPFFA(ours)	48.46

progressive feature fusion (PFF) and a multi-path feature attention mechanism (FAM), where PFF and FAM blocks are used in pairs. PFF can use the previous features to progressively guide and assist the subsequent feature learning, while FAM can weightedly concatenate the multi-path feature channels to improve the feature utilization and expressiveness. The experimental results show that our method is effective and significantly outperforms other state-of-the-art methods in both objective and subjective measures. Our method improves the balance of model complexity and performance of the lightweight SISR models to a new level.

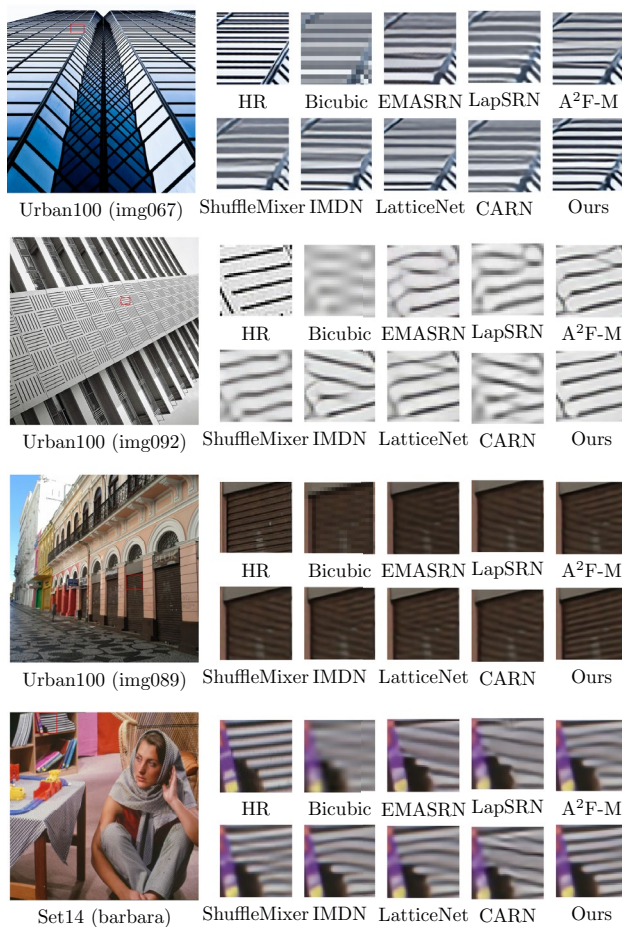


Fig. 3 Visual comparison for 4× SR

Data availability The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request. The data we used are all publicly available, and the sources of all datasets can be found in the references of the original article.

References

- Agarap A (2018) Deep learning using rectified linear units (relu). Preprint at [arXiv:1803.08375v2](https://arxiv.org/abs/1803.08375v2)
- Ahn N, Kang B, Sohn KA (2018) Fast, accurate, and lightweight super-resolution with cascading residual network. In: 15th European Conference on computer vision (ECCV). Springer, Munich, pp 256–272. https://doi.org/10.1007/978-3-030-01249-6_16
- Bevilacqua M, Roumy A, Guillemot C, et al (2012) Low-complexity single-image super-resolution based on nonnegative neighbor embedding. In: 23rd British Machine Vision Conference (BMVC). <https://doi.org/10.5244/C.26.135>
- Cao C, Liu X, Yang Y, et al (2015) Look and think twice: capturing top-down visual attention with feedback convolutional neural networks. In: IEEE International Conference on computer vision (ICCV). Springer, Santiago, pp 2956–2964. <https://doi.org/10.1109/ICCV.2015.338>
- Dai T, Cai JR, Zhang YB, et al (2019) Second-order attention network for single image super-resolution. In: IEEE/CVF Conference on computer vision and pattern recognition (CVPR). IEEE, pp 11057–11066. <https://doi.org/10.1109/CVPR.2019.01132>
- Dong C, Loy CC, He KM et al (2015) Image super-resolution using deep convolutional networks. *IEEE Trans Pattern Anal Mach Intell* 38(2):295–307. <https://doi.org/10.1109/TPAMI.2015.2439281>
- Dong C, Loy CC, Tang XO (2016) Accelerating the super-resolution convolutional neural network. In: 14th European Conference on computer vision (ECCV). Springer, Amsterdam, pp 391–407. https://doi.org/10.1007/978-3-319-46475-6_257
- Freeman WT, Pasztor EC, Carmichael OT (2000) Learning low-level vision. *Int J Comput Vis* 40(1):25–47. <https://doi.org/10.1109/ICCV.1999.790414>
- Gao H, Yuan H, Wang Z et al (2019) Pixel transposed convolutional networks. *IEEE Trans Pattern Anal Mach Intell* 42(5):1218–1227. <https://doi.org/10.1109/TPAMI.2019.2893965>
- Goodfellow I, Pouget-Abadie J, Mirza M, et al (2014) Generative adversarial nets. In: Ghahramani Z, Welling M, Cortes C, et al (eds) *Advances in neural information processing systems*, vol 27. Curran Associates, Inc. <https://proceedings.neurips.cc/paper/2014/file/5ca3e9b122f61f8f06494c97b1afccf3-Paper.pdf>. Accessed 17 Dec 2022
- Han W, Chang S, Liu D, et al (2018) Image super-resolution via dual-state recurrent networks. In: 2018 IEEE Conference on computer vision and pattern recognition. IEEE, pp 1654–1663. <https://doi.org/10.1109/CVPR.2018.00178>
- He K, Zhang X, Ren S, et al (2015) Delving deep into rectifiers: surpassing human-level performance on imagenet classification. In: 2015 IEEE International Conference on computer vision (ICCV). IEEE, pp 1026–1034. <https://doi.org/10.1109/ICCV.2015.123>
- Howard AG, Zhu M, Chen B, et al (2017) Mobilenets: efficient convolutional neural networks for mobile vision applications. Preprint at [arXiv:1704.04861](https://arxiv.org/abs/1704.04861)
- Hu J, Shen L, Sun G (2019) Squeeze-and-excitation networks. *IEEE Trans Pattern Anal Mach Intell* 42:2011–2023. <https://doi.org/10.1109/TPAMI.2019.2913372>. (IEEE)
- Huang G, Liu Z, Van Der Maaten L, et al (2017) Densely connected convolutional networks. In: 2017 IEEE Conference on computer vision and pattern recognition. IEEE, pp 2261–2269. <https://doi.org/10.1109/CVPR.2017.243>
- Huang JB, Singh A, Ahuja N (2015) Single image super-resolution from transformed self-exemplars. In: 2015 IEEE Conference on computer vision and pattern recognition. IEEE, pp 5197–5206. <https://doi.org/10.1109/CVPR.2015.7299156>
- Huang Z, Li W, Li J et al (2021) Dual-path attention network for single image super-resolution. *Expert Syst Appl* 169(114):450. <https://doi.org/10.1016/j.eswa.2020.114450>
- Hui Z, Wang XM, Gao XB (2018) Fast and accurate single image super-resolution via information distillation network. In: 2018 IEEE/CVF Conference on computer vision and pattern recognition. IEEE, pp 723–731. <https://doi.org/10.1109/CVPR.2018.00082>
- Hui Z, Gao X, Yang Y, et al (2019) Lightweight image super-resolution with information multi-distillation network. In: 27th ACM International Conference on multimedia. association for computing machinery, pp 2024–2032. <https://doi.org/10.1145/3343031.3351084>
- Kim J, Lee JK, Lee KM (2016) Accurate image super-resolution using very deep convolutional networks. In: 2016 IEEE Conference on computer vision and pattern recognition. IEEE, pp 1646–1654. <https://doi.org/10.1109/CVPR.2016.182>
- Kim J, Lee JK, Lee KM (2016) Deeply-recursive convolutional network for image super-resolution. In: 2016 IEEE Conference

- on computer vision and pattern recognition. IEEE, pp 1637–1645. <https://doi.org/10.1109/CVPR.2016.181>
22. Kim J, Lee JK, Lee KM (2017) Image super-resolution using dense skip connections. In: 2017 IEEE International Conference on computer vision. IEEE, pp 4799–4807. <https://doi.org/10.1109/CVPR.2016.182>
 23. Kingma D, Ba J (2014) Adam: a method for stochastic optimization. Comput Sci Preprint at [arXiv:1412.6980](https://arxiv.org/abs/1412.6980)
 24. Lai WS, Huang JB, Ahuja N, et al (2017) Deep Laplacian pyramid networks for fast and accurate super-resolution. In: 2017 IEEE Conference on computer vision and pattern recognition. IEEE, pp 5835–5843. <https://doi.org/10.1109/CVPR.2017.618>
 25. Ledig C, Theis L, Huszar F, et al (2017) Photo-realistic single image super-resolution using a generative adversarial network. In: Proceedings of the IEEE Conference on computer vision and pattern recognition (CVPR)
 26. Li W, Zhou K, Qi L, et al (2021) Lapa: inearily-assembled pixel-adaptive regression network for single image super-resolution and beyond. Adv Neural Inform Process Syst 33:20343–20355. Preprint at [arXiv:2105.10422](https://arxiv.org/abs/2105.10422)
 27. Lim B, Son S, Kim H, et al (2017) Enhanced deep residual networks for single image super-resolution. In: Proceedings of the IEEE Conference on computer vision and pattern recognition workshops. IEEE, Honolulu, pp 136–144. <https://doi.org/10.1109/CVPRW.2017.151>
 28. Liu J, Tang J, Wu G (2020) Residual feature distillation network for lightweight image super-resolution. In: Proceedings of the 16th European Conference on computer vision. Springer, Glasgow, pp 41–55. https://doi.org/10.1007/978-3-030-67070-2_2
 29. Liu JJ, Hou Q, Cheng MM, et al (2020) Improving convolutional networks with self-calibrated convolutions. In: 2020 IEEE/CVF Conference on computer vision and pattern recognition. IEEE, pp 10096–10105. <https://doi.org/10.1109/CVPR42600.2020.01011>
 30. Luo X, Xie Y, Zhang Y, et al (2020) Latticenet: towards lightweight image super-resolution with lattice block. In: 16th European Conference on computer vision. Springer, Glasgow, pp 272–289. https://doi.org/10.1007/978-3-030-58542-6_17
 31. Martin D, Fowlkes C, Tal D, et al (2001) A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In: 8th IEEE International Conference on computer vision, vol 2. IEEE, pp 416–423. <https://doi.org/10.1109/ICCV.2001.937655>
 32. Matsui Y, Ito K, Aramaki Y et al (2017) Single image super-resolution from transformed self-exemplars. Multimed Tools Appl 76(20):21811–21838. <https://doi.org/10.1007/s11042-016-4020-z>
 33. Muqeet A, Hwang J, Yang S, et al (2020) Multi-attention based ultra lightweight image super-resolution. In: 16th European Conference on computer vision. Springer, Glasgow, pp 103–118. https://doi.org/10.1007/978-3-030-67070-2_6
 34. Park SC, Park MK, Kang MG (2003) Super-resolution image reconstruction a technical overview. IEEE Signal Process Mag 20(3):21–36. <https://doi.org/10.1109/MSP.2003.1203207>
 35. Paszke A, Gross S, Chintala S et al (2017) Automatic differentiation in pytorch. 31st Conference on Neural Information Processing Systems (NIPS 2017), 4–9 December 2017, Long Beach, pp 1–4
 36. Peyré G, Bougleux S, Cohen L (2008) Non-local regularization of inverse problems. In: European Conference on computer vision. Springer, Marseille, pp 57–68. https://doi.org/10.1007/978-3-540-88690-7_5
 37. Shi WZ, Caballero J, Ledig C, et al (2013) Cardiac image super-resolution with global correspondence using multi-atlas patchmatch. In: International Conference on medical image computing and computer-assisted intervention. Springer, Nagoya, pp 9–16. https://doi.org/10.1007/978-3-642-40760-4_2
 38. Sun L, Pan J, Tang J (2022) Shufflemixer: an efficient convnet for image super-resolution. In: Advances in Neural Information Processing Systems, vol 35
 39. Tai Y, Liu XM, Xu CY (2017) Memnet: a persistent memory network for image restoration. In: 2017 IEEE International Conference on computer vision. IEEE, pp 4539–4547. <https://doi.org/10.1109/ICCV.2017.486>
 40. Tai Y, Yang J, Liu X (2017) Image super-resolution via deep recursive residual network. In: 2017 IEEE Conference on computer vision and pattern recognition. Springer, Munich, pp 3147–3155. <https://doi.org/10.1109/CVPR.2017.298>
 41. Thornton WM, Atkinson PM, Holland DA (2006) Sub-pixel mapping of rural land cover objects from fine spatial resolution satellite sensor imagery using super-resolution pixel-swapping. Int J Remote Sens 27(3):473–491. <https://doi.org/10.1080/01431160500207088>
 42. Timofte R, Agustsson E, Van Gool L, et al (2017) Ntire 2017 challenge on single image super-resolution: methods and results. In: 2017 IEEE Conference on computer vision and pattern recognition workshops. <https://doi.org/10.1109/CVPRW.2017.149>
 43. Wang C, Li Z, Shi J (2019) Lightweight image super-resolution with adaptive weighted learning network. Preprint at [arXiv:1904.02358](https://arxiv.org/abs/1904.02358)
 44. Wang X, Wang Q, Zhao Y, et al (2020) Lightweight single-image super-resolution network with attentive auxiliary feature learning. In: 15th Asian Conference on computer vision. Springer, Kyoto, pp 268–285. https://doi.org/10.1007/978-3-030-69532-3_17
 45. Wang XT, Yu K, Wu SX, et al (2018) Esrgan: enhanced super-resolution generative adversarial networks. In: 2018 European Conference on computer vision (ECCV). Springer, Munich, pp 63–79. https://doi.org/10.1007/978-3-030-11021-5_5
 46. Wang Z, Bovik AC, Sheikh HR et al (2004) Image quality assessment: from error visibility to structural similarity. IEEE Trans Image Process 13(4):600–612
 47. Woo S, Park JY, Lee JY, et al (2018) Cbam: convolutional block attention module. In: 2018 European conference on computer vision (ECCV). Springer, Munich, pp 3–19. https://doi.org/10.1007/978-3-030-01234-2_1
 48. Ymff0592 (2021) Super-resolution. <https://github.com/ymff0592/super-resolution>. Accessed 31 Mar 2022
 49. Yu J, Fan Y, Huang T (2019) Wide activation for efficient image and video super-resolution. In: Sidorov K, Hicks Y (eds) Proceedings of the British Machine Vision Conference (BMVC). BMVA Press, pp 52.1–52.13. <https://doi.org/10.5244/C.33.52>
 50. Ww Z, Yuen PC (2010) Very low resolution face recognition problem. Int J Remote Sens 21(1):327–340. <https://doi.org/10.1109/BTAS.2010.5634490>
 51. Zeyde R, Elad M, Protter M (2010) On single image scale-up using sparse-representations. In: 7th International Conference on curves and surfaces. Springer, Avignon, pp 711–730. https://doi.org/10.1007/978-3-642-27413-8_47
 52. Zhang J, Long C, Wang Y et al (2021) A two-stage attentive network for single image super-resolution. IEEE Trans Circ Syst Video Technol 32(3):1020–1033. <https://doi.org/10.1109/TCSVT.2021.3071191>
 53. Zhang K, Zuo W, Zhang L (2018) Learning a single convolutional super-resolution network for multiple degradations. In: 2018 IEEE/CVF Conference on computer vision and pattern recognition. IEEE, pp 3262–3271. <https://doi.org/10.1109/CVPR.2018.00344>
 54. Zhang X, Zhou X, Lin M, et al (2018) Shufflenet: an extremely efficient convolutional neural network for mobile devices. In:

- 2018 IEEE/CVF conference on computer vision and pattern recognition. IEEE, pp 6848–6856. <https://doi.org/10.1109/CVPR.2018.00716>
55. Zhang YL, Li KP, Li K, et al (2018) Image super-resolution using very deep residual channel attention networks. In: 15th European Conference on computer vision (ECCV). Springer, Munich, pp 294–310. https://doi.org/10.1007/978-3-030-01234-2_18
 56. Zhang YL, Li KP, Li K, et al (2018) Residual dense network for image super-resolution. In: 15th IEEE Conference on computer vision and pattern recognition. Springer, Munich, pp 294–310. https://doi.org/10.1007/978-3-030-01234-2_18
 57. Zhao H, Kong X, He J, et al (2020) Efficient image super-resolution using pixel attentio. In: 16th European conference on computer vision. Springer, Glasgow, pp 56–72, https://doi.org/10.1007/978-3-030-67070-2_3
 58. Zhao X, Liao Y, He T, et al (2019) F c² n: fully channel-concatenated network for single image super-resolution. Preprint at [arXiv:1907.03221v2](https://arxiv.org/abs/1907.03221v2)
 59. Zhou D, Duan R, Zhao L et al (2020) Single image super-resolution reconstruction based on multi-scale feature mapping adversarial network. *Signal Process* 166(107):251. <https://doi.org/10.1016/j.sigpro.2019.107251>
 60. Zhou D, Chen Y, Li W et al (2021) Image super-resolution based on adaptive cascading attention network. *Expert Syst Appl* 186(115):815. <https://doi.org/10.1016/j.eswa.2021.115815>
 61. Zhu X, Guo K, Ren S, et al (2022) Lightweight image super-resolution with expectation-maximization attention mechanism. *IEEE Transactions on Circuits and Systems for Video Technology* 32(3):1273C1284. <https://doi.org/10.1109/TCSVT.2021.3078436>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.